EE134: Fundamentals of Photovoltaic Devices

Spring 2026

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Abstract

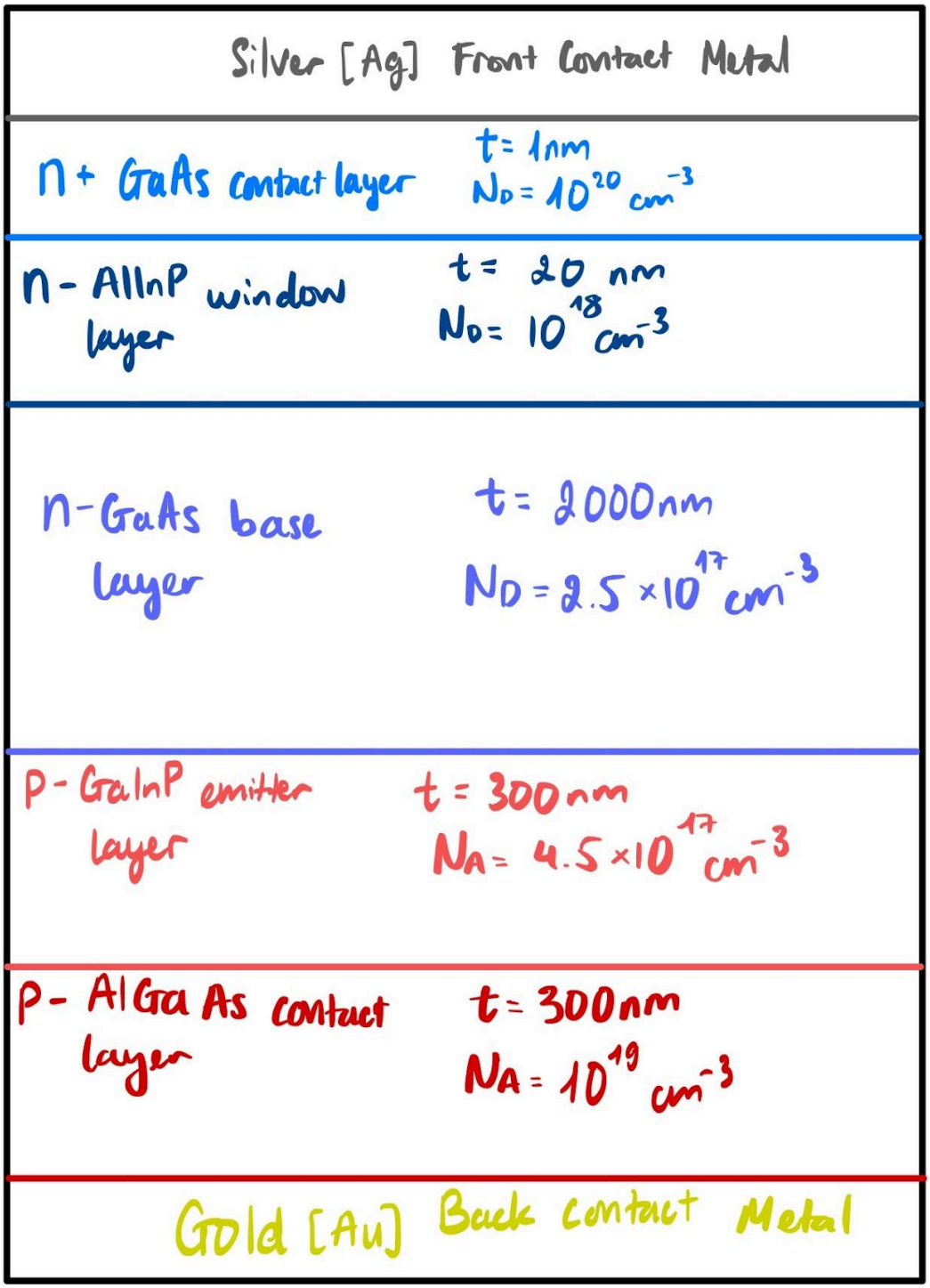
This project presents the design and optimization of a single-junction gallium arsenide (GaAs) heterojunction solar cell using AFORS-HET simulation. The device structure is based on a multilayer architecture incorporating n+ GaAs, AlInP window, GaAs base, GaInP emitter, and AlGaAs contact layers, with modified metal contacts to improve carrier extraction. In particular, the front contact material was changed to silver (Ag) to reduce Schottky barrier effects and enhance device performance. Parametric studies were conducted to evaluate the impact of base thickness and emitter doping on key performance metrics, including open-circuit voltage (Voc), short-circuit current density (Jsc), fill factor (FF), and efficiency. Results show that base thickness plays a dominant role, with an optimal range around 1750–1850 nm that balances light absorption and recombination losses. Emitter doping exhibits a weaker influence, with optimal performance in the mid-to-high doping range. The optimized device achieved a Voc of 1107 mV, Jsc of 25.78 mA/cm^2, FF of 87.73%, and an overall efficiency of 25.04%. While the simulated I–V characteristics show good qualitative agreement with literature, deviations arise due to modeling assumptions and simplified material parameters. This study highlights the importance of balancing competing physical mechanisms in photovoltaic device design and demonstrates the effectiveness of simulation-based optimization.

I. Introduction:

By directly converting sunlight into electricity, photovoltaic cells serve a critical role in modern energy systems, powering anything from satellites to residential homes. The photovoltaic effect, in which incoming light creates electron–hole pairs inside a semiconductor, is the basis for their operation. These charge carriers are subsequently separated by an internal electric field, creating a useful electrical current. Gallium arsenide (GaAs) stands out among the several materials used for PV devices because of its direct bandgap, high efficiency, and robust performance at high temperatures and radiation levels. Because of these characteristics, GaAs is particularly well-suited for harsh settings where dependability and efficiency are crucial, like space applications and cutting-edge solar-powered automobiles. For this project, I used Afors-Het to model a single-junction GaAs solar cell according to the construction outlined in Schulte et al.'s paper "Modeling and Design of III-V Heterojunction Solar Cells for Enhanced Performance" but there were some limitations due to the fact that some parameters were missing but I was able to assume them successfully to make it work.

II. Theory

A. Cell Structure and Layer Properties:



Above we have the cell structure diagram where the Front Contact Metal was chosen to be silver. In Schulte et al.'s paper "Modeling and Design of III-V Heterojunction Solar Cells for Enhanced Performance", the author didn't specify or talk about the front contact or the thickness of the n+GaAs contact layer so I decided to use [Ag] as the front for proper contact behavior.
The second layer will be n+ Gallium Arsenide [GaAs] Contact Layer where it is heavily n-type doped where it provides a highly conductive interface between the metal contact and the device and enables efficient electron injection/extraction at the front contact.

The third layer that was used is Aluminum Indium Phosphide [AlInP] where it is n-doped and being a thicker layer than the previous. AlInP's layer job is to reduce series resistance and improves electron extraction, acts as a surface passivation layer, and provides a wide bandgap barrier to block minority carriers from reaching the surface.

The fourth layer is the thickest layer in the structure, where it is an n-doped based layer of [GaAs] so that it can be the primary photon absorption region as well as generates electron–hole pairs across a broad solar spectrum. The doping level for this layer is not too high in order to reduce impurity scattering, and high crystalline quality is important to help minimize recombination losses.

The design began to shift to a p-type layer, where it introduced a p-doped GaInP emitter layer to help form the heterojunction using GaAs absorption as well as help move the depletion region into the emitter in order to help lower minority carrier concentration in the absorber, reduces SRH recombination as well as dark current.

Another p-type layer is added after using [AlGaAs] so that it could form a low-resistance ohmic contact and make the back surface stronger. Due to this layer having a high doping, it improves carrier extraction, reduces recombination at the rear surface where the conduction band offset acts as a barrier that redirects minority electrons toward the junction.

Lastly, the structure ends with gold [Au] as the Back Contact Metal where it forms a low-resistance ohmic interface with the p+ layer, enabling efficient hole extraction and minimizing resistive losses. The device uses a multi-layer heterojunction design that separates photon absorption and carrier collection, optimizing absorption in the GaAs layer while improving carrier extraction through tuned interfaces. Key features such as bandgap engineering, doping asymmetry, and a back-surface field work together to reduce recombination and control carrier flow, while graded interfaces smooth energy barriers. As a result, the cell achieves high performance, including strong open circuit voltage (Voc), high fill factor (FF), and efficient charge transport with minimized losses.

Note: All of the refractive indexes were manually inputted from refractiveindex.info.

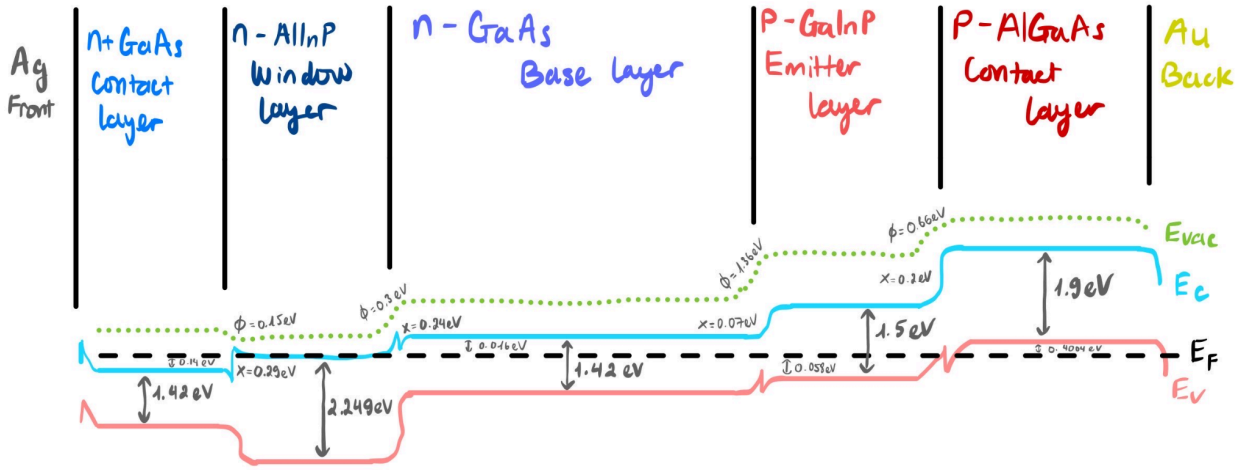
B. Front and Back Contact Design:

Looking at Schulte et al. 's design, I noticed that the front and back contact were designed the same but when Au is being directly on AllnP, it created a Schottky contact with a large barrier height meaning poor efficiency. So a thin, heavily n-doped GaAs layer was added between the metal and the window layer to reduce the work function mismatch, it did helped but the barrier height was still too large so I decided to switch to using a Ag as the front contact instead which has a workfunction of 4.2 eV which is closer to the electron affinity of GaAs being 4.07 eV. With this, it reduced the barrier height to about 0.13 eV. Although the contact is still Schottky, the smaller barrier significantly improved the cell efficiency to ~24%.

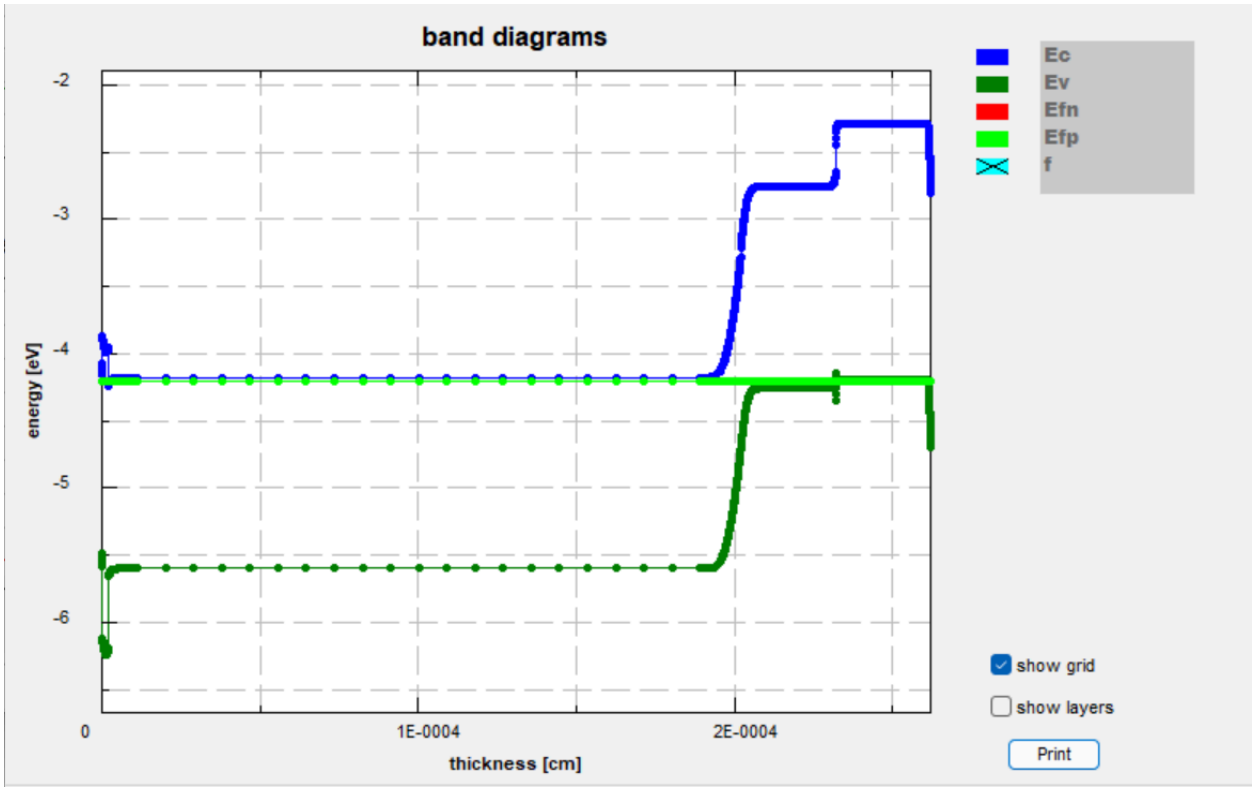
As for the back contact metal, I kept it the same as the original design to be Au with band bending turned on and metal work function being 5.2 eV (Au in room temperature) which resulted in Ohmic contact between the AlGaAs contact layer and the Au back contact metal (this would be best for photocurrent).

III. Design

A. Dark Energy Band Diagrams (Draw):

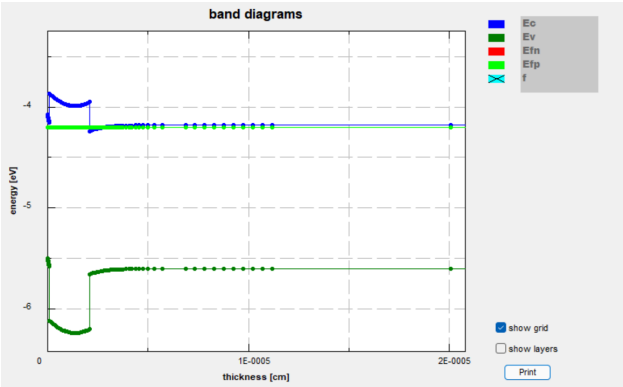


AFORS-HET Band Diagram Output:

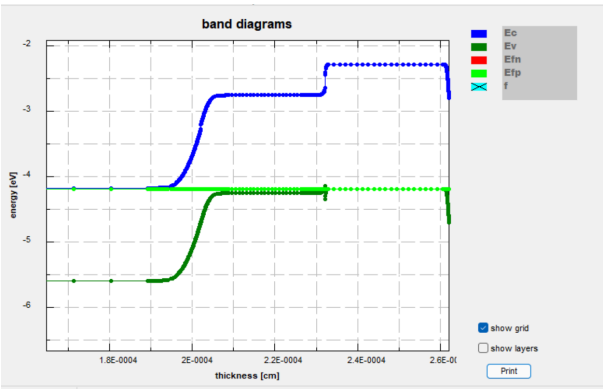


Overall Output

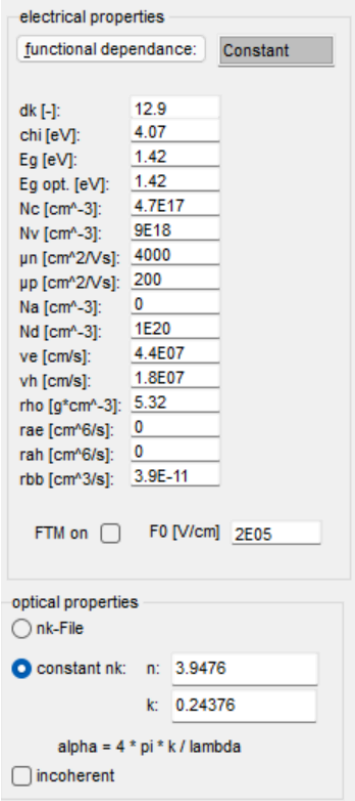
B. Parameters:



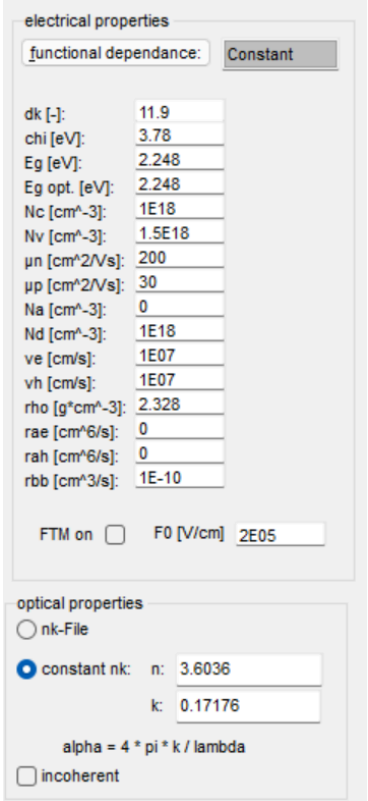
Front Contact



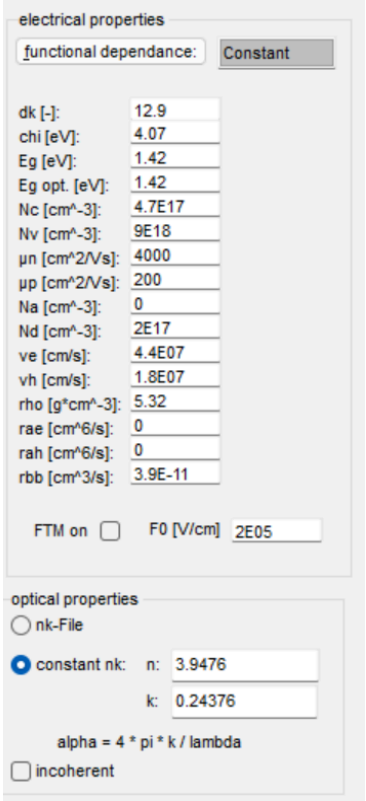
Back Contact



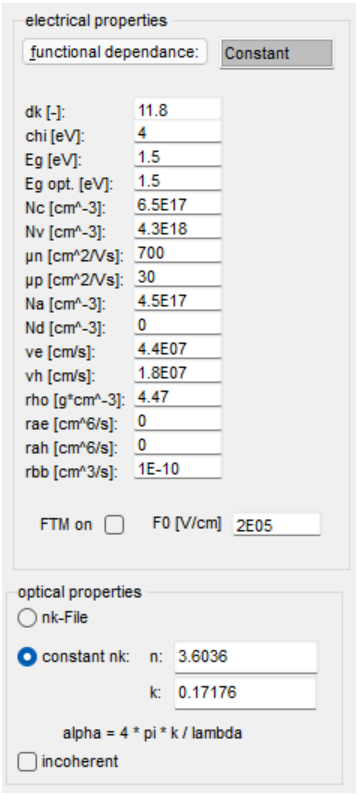
n+ GaAs (Contact)



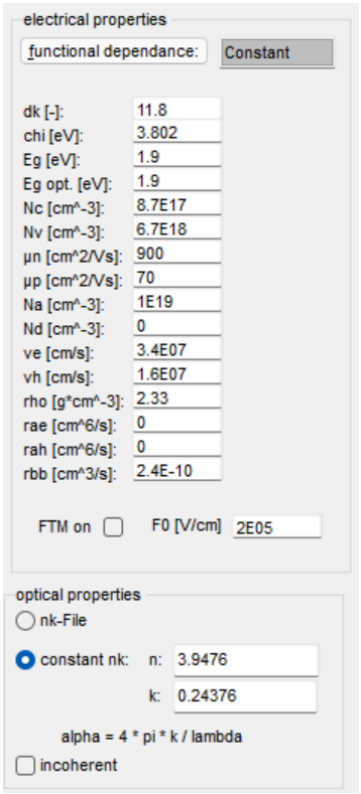
n-AlInP (Window)



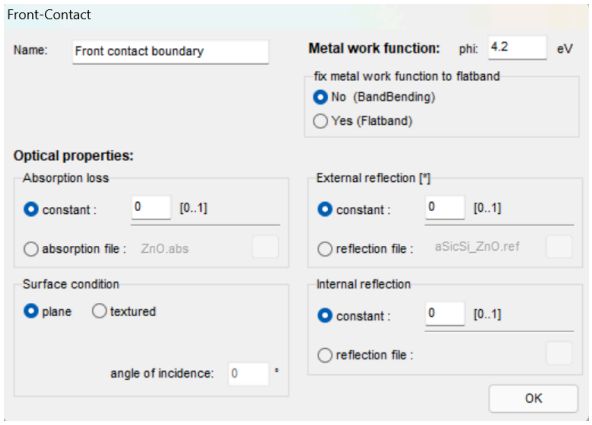
n-GaAs (Base)



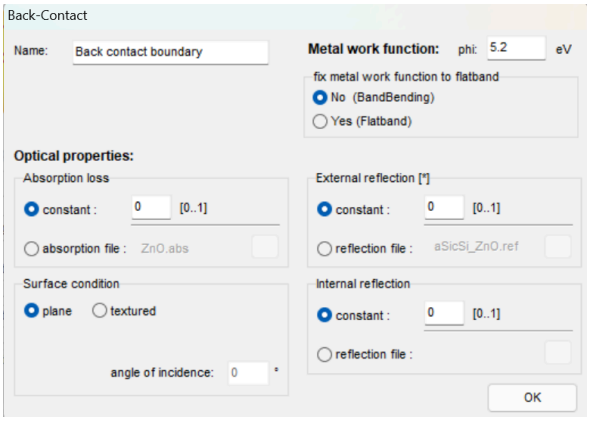
p-GaInP (Emitter)



p-AlGaAs (Contact)



Ag (Front Contact)



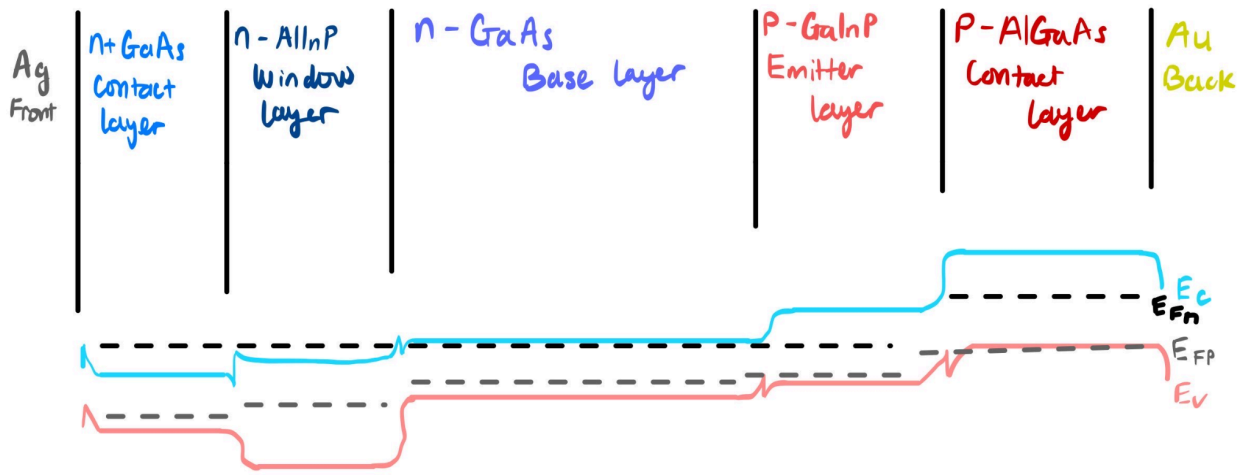
Au (Back Contact)

As we can see above,the band diagram of both the hand drawn and the AFORS-HET simulated are similar, bending mostly upwards as it enters the n-GaAs base layer, because this is mostly based on the set thicknesses of each material.

In the hand drawn diagram, we see that each layer is “thick enough” to behave like its own bulk material. That’s why I chose to put the Fermi level right on the conduction band in the window and n-doped layers. Whereas in AFORS-HET, the program simulated it as if they are almost entirely influenced by the interfaces next to them, making the layers so thin that it's being dominated by whatever is next to them.

There are some discontinuities in the band diagram in the case of heterojunctions. The calculated gaps are labeled as “X” with their respective values in the hand drawing, these gaps can be seen as a disadvantage due to the fact that they are actually restricting electrons and holes drift/diffusion.

Under Illumination



When the junction is under illumination, the single equilibrium Fermi level splits into two quasi-Fermi levels, E_{Fn} and E_{Fp}, as shown in the hand-drawn diagram above. This occurs because photogenerated electrons and holes are no longer in equilibrium. Electrons accumulate toward the n-side while holes accumulate toward the p-side, creating separate electrochemical potentials for each carrier type. The conduction band E_C and valence band E_V remain approximately unchanged in shape since they are determined by the material properties and built-in electric field. However, the quasi-Fermi levels are position-dependent and their separation is largest in the depletion and base regions where carrier generation is significant. Overall, illumination introduces non-equilibrium conditions, resulting in quasi-Fermi level splitting while preserving the general band structure of the device.

C. Optimization study:

I've collected data from the IV curves generated by the AFORS-HET using different parameters on base thickness and emitter doping, illuminated under AM1.5 (ASTMG173_AM15.in). The two tables summarize the results of Voc (mV), Jsc (mA/cm^2), FF (%), Eff (%) which shows some interesting trends. The blue highlights show decreasing trends while red highlights show an increasing trend. The green highlights will be used for the optimized design.

Varying Base Thickness

Thickness (nm)	Voc (mV)	Jsc (mA/cm^2)	FF (%)	Eff (%)
500	1134	22.75	87.80	23.02
750	1127	23.76	87.87	23.52
1000	1120	24.73	87.92	24.35
1250	1115	25.29	87.86	24.78
1500	1111	25.61	87.82	24.98
1750	1107	25.78	87.70	25.04
1800	1107	25.79	87.70	25.04
1850	1106	25.80	87.70	25.04
1900	1105	25.82	87.69	25.03
1950	1105	25.83	87.69	25.03
2000	1104	25.85	87.68	25.02
2050	1103	25.85	87.24	24.89
2100	1102	25.84	87.03	24.80
2250	1101	25.84	86.84	24.71
2500	1099	25.79	86.94	24.63
2750	1096	25.70	87.05	24.52
3000	1094	25.58	87.10	24.38
3250	1092	25.44	87.15	24.21
3500	1090	25.32	87.20	24.07

Varying Base Thickness: The effect of base thickness on solar cell performance was evaluated over a range from 500 nm to 3500 nm. The results show clear trade-offs between carrier generation and recombination. As base thickness increases, Jsc rises significantly at first, from 22.75 mA/cm^2 at 500 nm to approximately 25.85 mA/cm^2 near 2000 nm. This increase is expected, as a thicker base absorbs more incident light, generating more carriers. However, beyond ~2000 nm, Jsc begins to slightly decrease, indicating that additional thickness no longer improves carrier collection due to increased recombination losses. In contrast, Voc decreases monotonically with increasing thickness, from 1134 mV at 500 nm to 1090 mV at 3500 nm. This trend is consistent with increased recombination in thicker devices, which reduces the quasi-Fermi level splitting. The fill factor (FF) remains relatively stable around 87–88% across all thicknesses, with only a slight decline at larger thickness values. This suggests that resistive and transport-related losses are not strongly affected by thickness in this range. As a result of these competing effects, the efficiency (PCE) increases with thickness initially, reaching a maximum of approximately 25.04% between 1750 nm and 1850 nm, and then gradually decreases for thicker bases. This indicates the presence of an optimal thickness region, where the balance between light absorption and recombination is maximized. Overall, the results demonstrate that increasing base thickness improves performance up to a point, after which recombination losses dominate, leading to reduced device efficiency.

Varying Emitter Doping

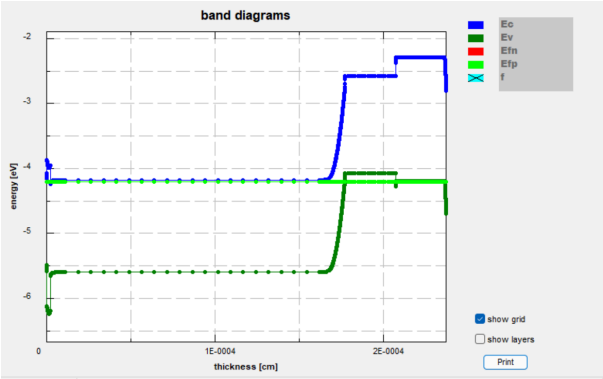
Doping (cm^{-3})	Voc (mV)	Jsc (mA/cm^2)	FF (%)	Eff (%)
4.5E10	1104	25.80	86.83	24.76
4.5E11	1104	25.81	86.95	24.78
4.5E12	1104	25.81	86.99	24.79
4.5E13	1104	25.81	86.99	24.79
4.5E14	1104	25.81	87.04	24.81
4.5E15	1104	25.82	87.35	24.90
4.5E16	1104	25.83	87.70	25.01
4.5E17	1104	25.85	87.68	25.02
4.5E18	1104	25.85	87.66	25.03
4.5E19	1104	25.85	87.66	25.03
4.5E20	1104	25.85	87.67	25.03
4.5E21	1104	25.85	87.59	25.01
4.5E22	1104	26.09	86.80	25.01
4.5E23	1104	26.25	86.62	25.00
4.5E24	1104	26.30	86.51	24.94

Varying Emitter Doping: The effect of emitter doping on solar cell performance was analyzed over a wide range from $4.5\times10^{10} cm^{-3}$ to $4.5\times10^{24} cm^{-3}$. The results show relatively weak dependence compared to base thickness, with only subtle performance variations. The open-circuit voltage (Voc) remains constant at 1104 mV across all doping levels, indicating that emitter doping in this range does not significantly impact the built-in potential or dominant recombination mechanisms in the device. The short-circuit current density (Jsc) shows a slight increase from 25.80 to 26.30 mA/cm^2 as doping increases. This gradual improvement may be attributed to enhanced carrier collection or reduced contact resistance at higher doping levels. However, the overall change is small, suggesting that Jsc is largely insensitive to emitter doping. The fill factor (FF) initially increases with doping, reaching a maximum of approximately 87.70% near $4.5\times10^{16} cm^{-3}$, and then gradually decreases at higher doping levels. This behavior reflects the trade-off between reduced series resistance at moderate doping and increased recombination losses at very high doping. As a result, the efficiency (PCE) increases slightly with doping, peaking at about 25.03% in the range of 4.5×10^{18} to $4.5\times10^{20} cm^{-3}$, before declining marginally at higher doping concentrations. Overall, the results indicate that emitter doping has a secondary effect on device

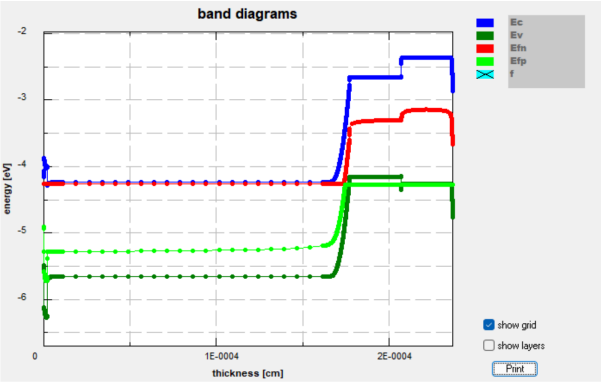
performance, with an optimal doping range in the mid-to-high 10^{16} to 10^{20} cm^{-3} region, where the balance between resistive losses and recombination is most favorable.

D. Optimized Design:

Thermal Equilibrium



Under Illumination

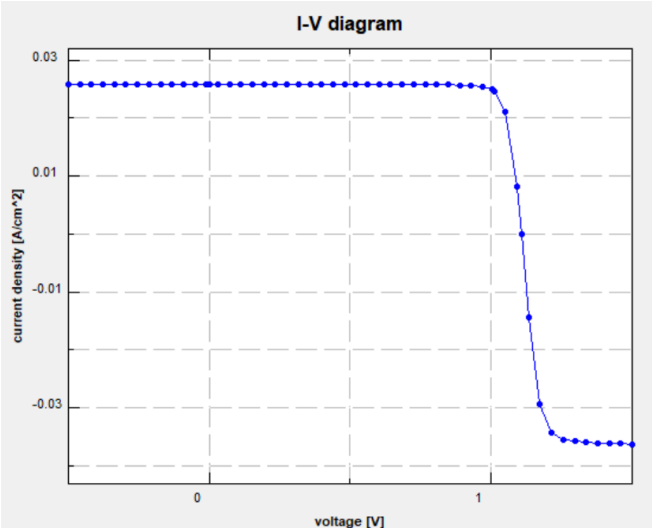


As we can see above, at thermal equilibrium, the band diagram shows a flat Fermi level across the full solar cell, which is expected because there is no external bias or illumination. The conduction band Ec and valence band Ev bend sharply near the junction, showing the built-in electric field created by the p-n junction. For the optimized structure with a base thickness of 1850 nm [Voc = 1106 mV, 25.80 mA/cm^2, 87.70%, 25.04%] and an emitter doping concentration of 4.50×10^{20} cm^{-3} [Voc = 1104 mV, 25.85 mA/cm^2, 87.67%, 25.03%], the band bending indicates that the device has a strong internal field to help separate carriers. Under illumination, the electron and hole quasi-Fermi levels split, which is expected because light generates excess carriers. This splitting represents the photovoltage of the solar cell. The conduction and valence bands keep a similar overall shape, but the separation between the quasi-Fermi levels shows that the device is producing electrical energy under light. Overall, the optimized base thickness and high emitter doping help create strong carrier separation while still allowing useful light absorption and current generation. With the combined optimized parameters, we were able to get Voc being 1107 mV, Jsc being 25.78 mA/cm^2, FF being 87.73%, and Eff being 25.04%.

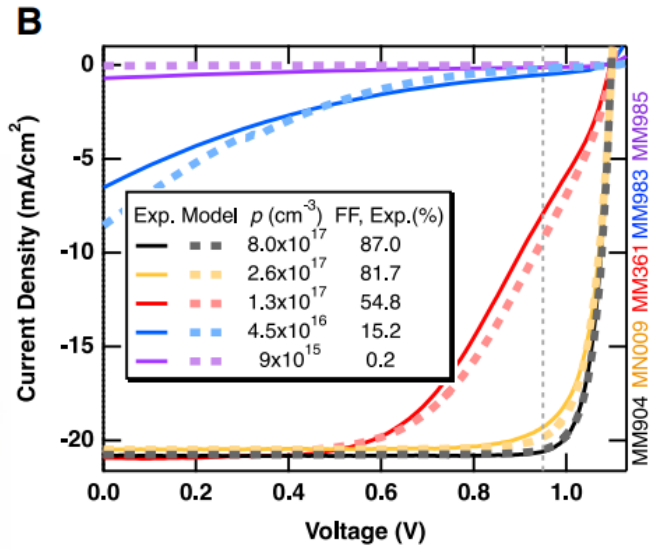
IV. Results

Voc = 1107 mV, Jsc = 25.78 mA/cm^2, FF = 87.73%, and Eff = 25.04%

AFORS-HET's Result

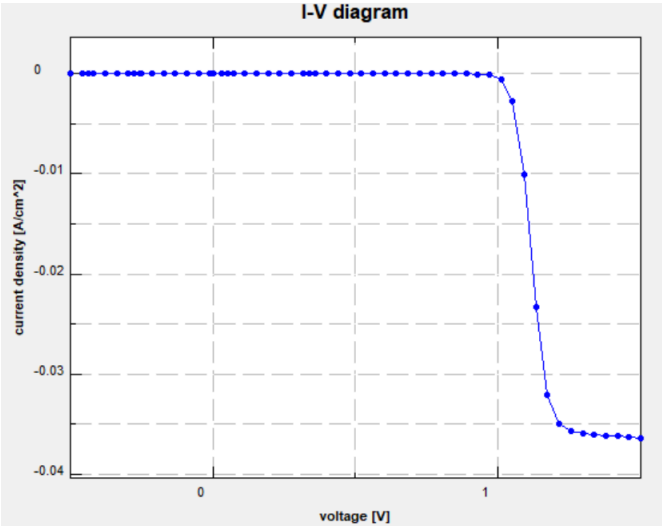


CellPress's Result



The simulated I-V curve from AFORS-HET shows good qualitative agreement with the GaAs heterojunction solar cell results reported in the CellPress's J-V curve, as both exhibit the expected flat short-circuit current region, a clear knee near the open-circuit voltage (~1 V), and a rapid drop in current beyond that point. However, the simulated curve appears more ideal, with a flatter current region and a sharper transition at the knee, indicating a higher fill factor than the experimental curves. These differences likely arise from me changing the front contact metal to being silver [Ag], an additional layer of n+GaAs contact layer was added as an attempt to mitigate any differences as well as refractive indexes were all used based on assumptions (no .nk files). As a result, while the simulation captures the correct physical trends, it does not fully reproduce the non-ideal effects observed in real devices, and therefore the agreement is qualitative rather than exact.

Dark I-V curve



Series resistance: $(V_1, I_1) = (1.15, -0.016)$ } Steep region points
$(V_2, I_2) = (1.20, -0.031)$ }

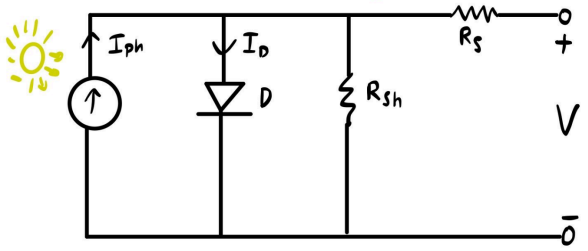
$R_S = \left|\frac{\Delta V}{\Delta I}\right| = \left|\frac{1.20 - 1.15}{-0.031 - (-0.016)}\right| = 3.\overline{3}\ \Omega \cdot cm^2$

Shunt resistance: $(V_1, I_1) \approx (0.000173842, -3.15083148 \times 10^{-4})$ } Flat region points
$(V_2, I_2) \approx (0.000466910\overline{35}, -3.15077927 \times 10^{-4})$ }

$R_{Sh} = \left|\frac{\Delta V}{\Delta I}\right| \approx 64368.18581 = 64.368k\ \Omega \cdot cm^2$

Diode ideally factor: $(V_1, J_1) \approx (1.0628, -6.37 \times 10^{-4})$ } Middle region points
$(V_2, J_2) \approx (1.1489, -1.85 \times 10^{-3})$ }
↳ A/cm²

$n = \frac{\Delta V}{V_T \ln\left(\frac{|J_2|}{|J_1|}\right)} = \frac{1.1489 - 1.0628}{0.02585 \ln\left(\frac{-1.85 \times 10^{-3}}{-6.37 \times 10^{-4}}\right)} = 3.12403$



The calculated series resistance, shunt resistance and diode ideally factor were all calculated through using the dark I-V curve slopes.

V. Conclusion

This project focused on designing and optimizing a GaAs heterojunction solar cell using AFORS-HET to understand how structural and material parameters influence device performance. Key design choices, including the multilayer heterostructure and the use of Ag as the front contact instead of the proposed material in the original design, improved carrier extraction and reduced barrier-related losses, enabling strong overall device performance.

The optimization results show that base thickness plays the most significant role. Increasing thickness improves light absorption and Jsc, but also increases recombination, leading to a steady decrease in Voc. This trade-off results in an optimal thickness around 1750–1850 nm, where efficiency peaks at approximately 25.04%. In contrast, emitter doping has a weaker influence, mainly affecting fill factor and slightly improving current collection at moderate-to-high doping levels, with an optimal range in the mid-to-high 10^{16} to 10^{20} cm^{-3} region.

The final optimized device achieved Voc = 1107 mV, Jsc = 25.78 mA/cm^2, FF = 87.73%, and efficiency = 25.04%, which is consistent with high-performance GaAs solar cells. The simulated I-V curve shows good qualitative agreement with literature, capturing the expected shape and behavior, though it appears more ideal due to simplified assumptions, change of design (front contact was changed and n+GaAs contact layer was added) and assumed .nk constant values instead since CellPress did not cite those.

Through this work, an important takeaway is how strongly device performance depends on balancing competing physical mechanisms—absorption versus recombination, and conductivity versus carrier lifetime. Additionally, extracting parameters such as series resistance, shunt resistance, and diode ideality factor from the dark I-V curve provided useful insight into internal loss mechanisms and how they relate to overall device efficiency.

If given more time, this study could be extended by incorporating wavelength-dependent optical data, modeling interface and surface recombination more accurately, and refining contact models to better capture Schottky and ohmic behavior. Further optimization could also explore additional parameters such as window layer thickness, graded bandgap structures, or multi-junction designs. Finally, comparing the simulation more directly with experimental data would help validate the model and improve its predictive accuracy.

VI. References:

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